UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A2	TRUE	Sampling converts a continuous sound wave into digital values.	Sampling measures the wave at intervals.
A3	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A4	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A5	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A6	FALSE	MIDI files store digitized audio samples.	MIDI stores musical instructions, not samples.
A7	TRUE	Hertz measures sound frequency.	Hertz is cycles per second.
A8	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A9	TRUE	Card-based authoring tools are best for messages with a clear beginning and end.	Cards support sequential navigation.
A10	TRUE	Cel animation is created by drawing each frame by hand.	Cel animation uses hand-drawn frames.
A11	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A12	TRUE	Tweening creates intermediate frames between keyframes.	Tweening fills the gaps.
A13	TRUE	RGB is an additive colour model.	RGB adds red, green and blue light.
A14	TRUE	A web browser displays hypertext documents.	Browsers render web pages.
A15	TRUE	Path animation moves an object along a predefined route.	Path animation follows a motion path.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	K	Keyframe	important frame in an animation
B2	H	Multimedia	integration of text, graphics, audio, video and animation
B3	O	Bitmap	image made of pixels
B4	E	Tweening	generating in-between frames
B5	J	Scanner	input device that digitises documents
B6	A	Storyboard	visual sequence plan
B7	G	Plotter	output device for large drawings
B8	B	Sampling	converting sound into digital values
B9	F	Vector	image made of mathematical objects
B10	L	Linear multimedia	presentation without user control of sequence
B11	M	Frame rate	frames displayed per second
B12	D	MIDI	standard for musical notes and instructions
B13	C	Card-based tool	authoring tool organised like pages or cards
B14	N	Serif	font with small strokes at character ends
B15	I	Chroma key	replacing a colour background

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	A	In-between frames
C2	A	Messages with a beginning and end
C3	A	A simulated environment
C4	A	It combines visuals and sound
C5	C	Structure web content
C6	A	CMYK
C7	B	Multimedia with hyperlinks
C8	B	A video compression family
C9	C	A visual sequence plan
C10	B	The user can control content and flow
C11	D	Permission to use a work
C12	A	Printed output
C13	A	Musical notes and instructions
C14	A	Compiler
C15	A	Non-linear
C16	D	Display web pages
C17	C	Replace a colour background
C18	B	Another resource
C19	C	Small strokes at character ends
C20	D	Merging of media and computing
C21	D	Scanner
C22	A	Pixels
C23	A	Plotter
C24	D	Vector
C25	A	Space between individual character pairs
C26	B	Play while downloading
C27	B	Intended users
C28	B	Automatic device recognition
C29	B	Along a predefined route
C30	A	Smooth jagged edges

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PART D - WORKED ANSWERS

QUESTION 1

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(5 marks)

QUESTION 2

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 3

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(2 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(4 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

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QUESTION 4

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

END OF MARKING SCHEME

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